Solution: To apply the boundary conditions at the plane between the two media, we first need to obtain an expression for the unit vector normal to the boundary. To apply the boundary condition given by Eq. (5.84), we need an expression for $\hat{\mathbf{n}}_2$, which isthe unit vector pointing away from medium 2 (**Fig. 5-25**).

From the geometry of the small triangle in **Fig. 5-25**,

$$\theta = \tan^{-1}\left(\frac{1}{2}\right) = 26.57^\circ.$$

Hence,

$$\hat{\mathbf{n}}_2 = \hat{\mathbf{x}}\cos\theta + \hat{\mathbf{y}}\sin\theta = \hat{\mathbf{x}}0.89 + \hat{\mathbf{y}}0.45.$$

Magnetic field $\mathbf{B}_1$ has a normal component $\mathbf{B}_{1n}$ along $\hat{\mathbf{n}}_2$ and a tangential component $\mathbf{B}_{1t}$:

$$\mathbf{B}_1 = \mathbf{B}_{1n} + \mathbf{B}_{1t}$$

with

$$\mathbf{B}_{1n} = B_{1n}\hat{\mathbf{n}}_2$$

and

$$B_{1n} = \hat{\mathbf{n}}_2 \cdot \mathbf{B}_1 = (\hat{\mathbf{x}}\cos\theta + \hat{\mathbf{y}}\sin\theta)\cdot(\hat{\mathbf{x}}2 + \hat{\mathbf{y}}3) = 2\cos\theta + 3\sin\theta.$$

The tangential component of $\mathbf{B}_1$ is

$$\begin{aligned}\mathbf{B}_{1t} &= \mathbf{B}_1 - \mathbf{B}_{1n} = \mathbf{B}_1 - B_{1n}\hat{\mathbf{n}}_2 \\ &= (\hat{\mathbf{x}}2 + \hat{\mathbf{y}}3) - (2\cos\theta + 3\sin\theta)(\hat{\mathbf{x}}\cos\theta + \hat{\mathbf{y}}\sin\theta) \\ &= \hat{\mathbf{x}}(2 - 2\cos^2\theta - 3\sin\theta\cos\theta) \\ &\quad + \hat{\mathbf{y}}(3 - 2\cos\theta\sin\theta - 3\sin^2\theta).\end{aligned}$$

Boundary conditions require that

$$\mathbf{B}_{1n} = \mathbf{B}_{2n}$$

and

$$\frac{\mathbf{B}_{1t}}{\mu_1} = \frac{\mathbf{B}_{2t}}{\mu_2},$$

which leads to

$$\begin{aligned}\mathbf{B}_2 = \mathbf{B}_{2n} + \mathbf{B}_{2t} &= \mathbf{B}_{1n} + \frac{\mu_2}{\mu_1}\mathbf{B}_{1t} \\ &= B_{1n}\hat{\mathbf{n}}_2 + \frac{\mu_2}{\mu_1}\mathbf{B}_{1t} \\ &= (2\cos\theta + 3\sin\theta)(\hat{\mathbf{x}}\cos\theta + \hat{\mathbf{y}}\sin\theta) \\ &\quad + \frac{\mu_2}{\mu_1}[\hat{\mathbf{x}}(2 - 2\cos^2\theta - 3\sin\theta\cos\theta) \\ &\quad + \hat{\mathbf{y}}(3 - 2\cos\theta\sin\theta - 3\sin^2\theta)] \\ &= \hat{\mathbf{x}}\left[2\cos^2\theta + 3\sin\theta\cos\theta + \frac{\mu_2}{\mu_1}(2 - 2\cos^2\theta - 3\sin\theta\cos\theta)\right] \\ &\quad + \hat{\mathbf{y}}\left[2\cos\theta\sin\theta + 3\sin^2\theta + \frac{\mu_2}{\mu_1}(3 - 2\cos\theta\sin\theta - 3\sin^2\theta)\right].\end{aligned}$$

For $\theta = 26.57^\circ$ and $\mu_2 = 2\mu_1$,

$$\mathbf{B}_2 = \hat{\mathbf{x}}1.2 + \hat{\mathbf{y}}4.6 \quad \text{(T)}.$$

Exercise 5-13: With reference to **Fig. 5-24**, determine the angle between $\mathbf{H}_1$ and $\hat{\mathbf{n}}_2 = \hat{\mathbf{z}}$ if $\mathbf{H}_2 = (\hat{\mathbf{x}}3 + \hat{\mathbf{z}}2)$ (A/m), $\mu_{r_1} = 2$, $\mu_{r_2} = 8$, and $\mathbf{J}_s = 0$.

Answer: $\theta = 20.6^\circ$. (See EM.)

5-7 Inductance

An inductor is the magnetic analogue of an electric capacitor. Just as a capacitor can store energy in the electric field in the medium between its conducting surfaces, an inductor can store energy in the magnetic field near its current-carrying conductors. A typical inductor consists of multiple turns of wire helically coiled around a cylindrical core (**Fig. 5-26(a)**). Such a structure is called a ***solenoid***. Its core may be air filled or may contain a magnetic material with magnetic permeability μ. If the wire carries a current I and the turns are closely spaced, the solenoid will produce a relatively uniform magnetic field within its interior with magnetic field lines resembling those of the permanent magnet (**Fig. 5-26(b)**).

5-7.1 Magnetic Field in a Solenoid

As a prelude to our discussion of inductance, we derive an expression for the magnetic flux density $\mathbf{B}$ in the interior

解决方案：为了在两种介质之间的平面上应用边界条件，我们首先需要获得垂直于边界的单位向量的表达式。应用方程给出的边界条件。(5.84)，我们需要一个 $\hat{n}_2$ 的表达式，它是指向远离介质 2 的单位向量（图 5-25）。

由图5-25中小三角形的几何形状可知，

$$\theta = \tan^{-1}\left(\frac{1}{2}\right) = 26.57^\circ.$$

因此，

$$\hat{\mathbf{n}}_2 = \hat{\mathbf{x}}\cos\theta + \hat{\mathbf{y}}\sin\theta = \hat{\mathbf{x}}0.89 + \hat{\mathbf{y}}0.45.$$

磁场 B_1 具有沿 $\hat{n}_2$ 的法向分量 B_{1n} 和切向分量 B_{1t}：

$$\mathbf{B}_1 = \mathbf{B}_{1n} + \mathbf{B}_{1t}$$

和

$$\mathbf{B}_{1n} = B_{1n}\hat{\mathbf{n}}_2$$

和

$$B_{1n} = \hat{\mathbf{n}}_2 \cdot \mathbf{B}_1 = (\hat{\mathbf{x}}\cos\theta + \hat{\mathbf{y}}\sin\theta)\cdot(\hat{\mathbf{x}}2 + \hat{\mathbf{y}}3) = 2\cos\theta + 3\sin\theta.$$

B_1 的切向分量为

$$\begin{aligned}\mathbf{B}_{1t} &= \mathbf{B}_1 - \mathbf{B}_{1n} = \mathbf{B}_1 - B_{1n}\hat{\mathbf{n}}_2 \\ &= (\hat{\mathbf{x}}2 + \hat{\mathbf{y}}3) - (2\cos\theta + 3\sin\theta)(\hat{\mathbf{x}}\cos\theta + \hat{\mathbf{y}}\sin\theta) \\ &= \hat{\mathbf{x}}(2 - 2\cos^2\theta - 3\sin\theta\cos\theta) \\ &\quad + \hat{\mathbf{y}}(3 - 2\cos\theta\sin\theta - 3\sin^2\theta).\end{aligned}$$

边界条件要求

$$\mathbf{B}_{1n} = \mathbf{B}_{2n}$$

和

$$\frac{\mathbf{B}_{1t}}{\mu_1} = \frac{\mathbf{B}_{2t}}{\mu_2},$$

这导致

$$\begin{aligned}\mathbf{B}_2 = \mathbf{B}_{2n} + \mathbf{B}_{2t} &= \mathbf{B}_{1n} + \frac{\mu_2}{\mu_1}\mathbf{B}_{1t} \\ &= B_{1n}\hat{\mathbf{n}}_2 + \frac{\mu_2}{\mu_1}\mathbf{B}_{1t} \\ &= (2\cos\theta + 3\sin\theta)(\hat{\mathbf{x}}\cos\theta + \hat{\mathbf{y}}\sin\theta) \\ &\quad + \frac{\mu_2}{\mu_1}[\hat{\mathbf{x}}(2 - 2\cos^2\theta - 3\sin\theta\cos\theta) \\ &\quad + \hat{\mathbf{y}}(3 - 2\cos\theta\sin\theta - 3\sin^2\theta)] \\ &= \hat{\mathbf{x}}\left[2\cos^2\theta + 3\sin\theta\cos\theta + \frac{\mu_2}{\mu_1}(2 - 2\cos^2\theta - 3\sin\theta\cos\theta)\right] \\ &\quad + \hat{\mathbf{y}}\left[2\cos\theta\sin\theta + 3\sin^2\theta + \frac{\mu_2}{\mu_1}(3 - 2\cos\theta\sin\theta - 3\sin^2\theta)\right].\end{aligned}$$

对于 $\theta = 26.57^\circ$ 和 $\mu_2 = 2\mu_1$，

$$\mathbf{B}_2 = \hat{\mathbf{x}}1.2 + \hat{\mathbf{y}}4.6 \quad \text{(T)}.$$

练习 5-13：参考图 5-24，如果 $H_2 = (\hat{x}3 + \hat{z}2)$ (A/m)、$\mu_{r_1} = 2$、$\mu_{r_2} = 8$ 和 $J_s = 0$，求 H_1 和 $\hat{n}_2 = \hat{z}$ 之间的角度。

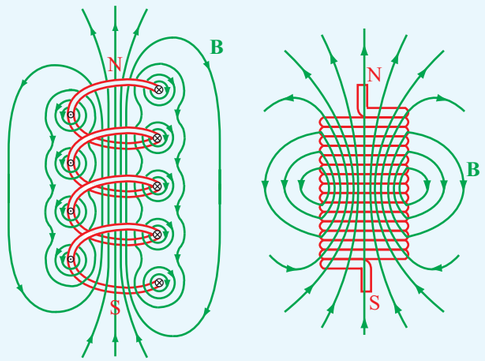
Answer: $\theta = 20.6^\circ$. (See EM.)

5-7 电感

电感器是电容器的磁性类似物。正如电容器可以在其导电表面之间的介质中的电场中存储能量一样，电感器可以在其载流导体附近的磁场中存储能量。典型的电感器由多匝螺旋状缠绕在圆柱形磁芯上的导线组成（图 5-26(a)）。这种结构称为螺线管。其磁芯可以是空气填充的，也可以包含具有磁导率 μ 的磁性材料。如果导线中流过电流I并且匝数间隔很近，螺线管将在其内部产生相对均匀的磁场，其磁场线类似于永磁体的磁场线（图5-26（b））。

5-7.1 螺线管中的磁场

作为讨论电感的前奏，我们推导了内部磁通密度 B 的表达式



(a) Loosely wound solenoid (b) Tightly wound solenoid

Figure 5-26 Magnetic field lines of (a) a loosely wound solenoid and (b) a tightly wound solenoid.

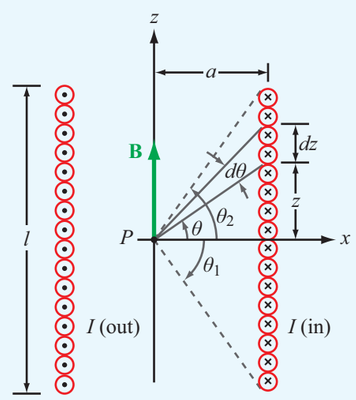
region of a tightly wound solenoid. The solenoid is of length l and radius a and comprises N turns carrying current I. The number of turns per unit length is $n = N/l$, and the fact that the turns are tightly wound implies that the pitch of a single turn is small compared with the solenoid's radius. Even though the turns are slightly helical in shape, we can treat them as circular loops (**Fig. 5-27(a)**). Let us start by considering the magnetic flux density **B** at point P on the axis of the solenoid. In Example 5-3, we derived the following expression for the magnetic field **H** along the axis of a circular loop of radius a, which is a distance z away from its center:

$$\mathbf{H} = \hat{\mathbf{z}}\,\frac{I'a^2}{2(a^2+z^2)^{3/2}}, \tag{5.86}$$

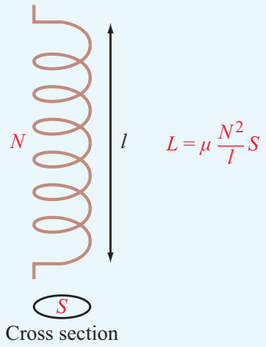
where I' is the current carried by the loop. If we treat an incremental length dz of the solenoid as an equivalent loop composed of $n\,dz$ turns carrying a current $I' = In\,dz$, then the induced field at point P is

$$d\mathbf{B} = \mu\,d\mathbf{H} = \hat{\mathbf{z}}\,\frac{\mu nIa^2}{2(a^2+z^2)^{3/2}}\,dz. \tag{5.87}$$

The total field **B** at P is obtained by integrating the contributions from the entire length of the solenoid. This is facilitated



(a) Cross section



Cross section

(b) Solenoid inductance

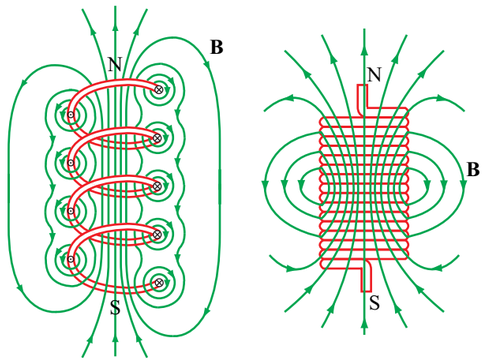
Figure 5-27 (a) Solenoid cross section showing geometry for calculating **H** at a point P on the solenoid axis and (b) solenoid inductance.

by expressing the variable z in terms of the angle θ, as seen from P to a point on the solenoid rim. That is,

$$z = a\tan\theta, \tag{5.88a}$$

$$a^2 + z^2 = a^2 + a^2\tan^2\theta = a^2\sec^2\theta, \tag{5.88b}$$

$$dz = a\sec^2\theta\,d\theta. \tag{5.88c}$$



(a) Loosely wound solenoid (b) Tightly wound solenoid

图 5-26 (a) 松散缠绕的螺线管和 (b) 紧密缠绕的螺线管的磁场线。

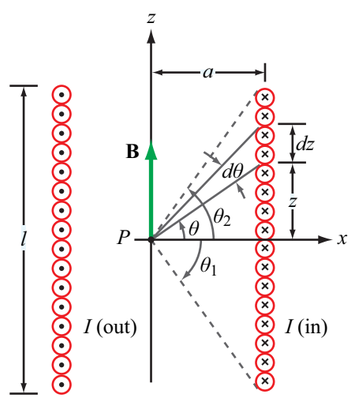
紧密缠绕的螺线管的区域。螺线管的长度为 l，半径为 a，包括承载电流 I 的 N 匝。每单位长度的匝数为 n = N/l，并且匝紧密缠绕的事实意味着单匝的节距与螺线管的半径相比较小。尽管匝的形状略呈螺旋状，但我们可以将它们视为圆形环（图5-27（a））。让我们首先考虑螺线管轴上 P 点的磁通密度 B。在例 5-3 中，我们推导出了沿半径为 a 的圆环轴的磁场 H 的以下表达式，圆环距圆心的距离为 z：

$$\mathbf{H} = \hat{\mathbf{z}}\,\frac{I'a^2}{2(a^2+z^2)^{3/2}}, \tag{5.86}$$

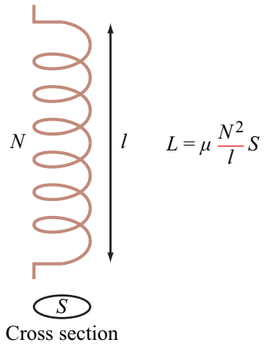
其中 I′ 是环路承载的电流。如果我们将螺线管的增量长度 dz 视为由 n dz 匝组成的等效回路，承载电流 I′ = In d z，则 P 点处的感应场为

$$d\mathbf{B} = \mu\,d\mathbf{H} = \hat{\mathbf{z}}\,\frac{\mu nIa^2}{2(a^2+z^2)^{3/2}}\,dz. \tag{5.87}$$

P 处的总磁场 B 是通过对螺线管整个长度的贡献进行积分而获得的。这有利于



(a) Cross section



Cross section

(b) Solenoid inductance

图 5-27 (a) 螺线管横截面显示了用于计算螺线管轴上点 P 处的 H 的几何形状，(b) 螺线管电感。

通过用角度 θ 表达变量 z，如从 P 到螺线管边缘上的点所见。那是，

$$z = a\tan\theta, \tag{5.88a}$$

$$a^2 + z^2 = a^2 + a^2\tan^2\theta = a^2\sec^2\theta, \tag{5.88b}$$

$$dz = a\sec^2\theta\,d\theta. \tag{5.88c}$$

Upon substituting the last two expressions in Eq. (5.87) and integrating from θ_1 to θ_2, we have

$$\mathbf{B} = \hat{\mathbf{z}}\,\frac{\mu nIa^2}{2}\int_{\theta_1}^{\theta_2}\frac{a\sec^2\theta\,d\theta}{a^3\sec^3\theta} = \hat{\mathbf{z}}\,\frac{\mu nI}{2}(\sin\theta_2 - \sin\theta_1). \tag{5.89}$$

If the solenoid length l is much larger than its radius a, then for points P away from the solenoid's ends $\theta_1 \approx -90^\circ$ and $\theta_2 \approx 90^\circ$, in which case Eq. (5.89) reduces to

$$\mathbf{B} \approx \hat{\mathbf{z}}\mu nI = \frac{\hat{\mathbf{z}}\mu NI}{l} \quad \text{(long solenoid with } l/a \gg 1\text{)}. \tag{5.90}$$

Even though Eq. (5.90) was derived for the field **B** at the midpoint of the solenoid, it is approximately valid everywhere in the solenoid's interior, except near the ends.

We now return to a discussion of inductance, which includes the notion of ***self-inductance***, representing the magnetic flux linkage of a coil or circuit with itself, and ***mutual inductance***, which involves the magnetic flux linkage in a circuit due to the magnetic field generated by a current in another one. Usually, when the term ***inductance*** is used, the intended reference is to self-inductance.

Exercise 5-14: Use Eq. (5.89) to obtain an expression for **B** at a point on the axis of a very long solenoid but situated at its end points. How does **B** at the end points compare to **B** at the midpoint of the solenoid?

Answer: $\mathbf{B} = \hat{\mathbf{z}}(\mu NI/2l)$ at the end points, which is half as large as **B** at the midpoint. (See EM.)

5-7.2 Self-Inductance of a Solenoid

From Eq. (5.66), the magnetic flux Φ linking a surface S is

$$\Phi = \int_S \mathbf{B}\cdot d\mathbf{s} \qquad \text{(Wb)}. \tag{5.91}$$

In a solenoid characterized by an approximately uniform magnetic field throughout its cross-section and given by Eq. (5.90), the flux linking a single loop is

$$\Phi = \int_S \hat{\mathbf{z}}\left(\mu\frac{N}{l}I\right)\cdot\hat{\mathbf{z}}\,ds = \mu\frac{N}{l}IS, \tag{5.92}$$

where S is the cross-sectional area of the loop (**Fig. 5-27(b)**). ***Magnetic flux linkage*** Λ is defined as the total magnetic flux linking a given circuit or conducting structure. If the structure consists of a single conductor with multiple loops, as in the case of the solenoid, Λ equals the flux linking all loops of the structure. For a solenoid with N turns,

$$\Lambda = N\Phi = \mu\frac{N^2}{l}IS \qquad \text{(Wb)}. \tag{5.93}$$

The ***self-inductance*** of any conducting structure is defined as the ratio of the magnetic flux linkage Λ to the current I flowing through the structure:

$$L = \frac{\Lambda}{I} \qquad \text{(H)}. \tag{5.94}$$

The SI unit for inductance is the henry (H), which is equivalent to webers per ampere (Wb/A).

For a solenoid, use of Eq. (5.93) gives

$$L = \mu\frac{N^2}{l}S \qquad \textbf{(solenoid)}. \tag{5.95}$$

5-7.3 Self-Inductance of Other Conductors

The solenoid consists of a single inductor of cross section S. If, on the other hand, the structure consists of two *separate* conductors, as in the case of the parallel-wire and coaxial transmission lines shown in **Fig. 5-28**, the flux linkage Λ associated with a length l of either line refers to the flux Φ through a closed surface between the two conductors, such as the shaded areas in **Fig. 5-28**. In reality, there is also some magnetic flux that passes through the conductors themselves, but it may be ignored by assuming that currents flow only on the surfaces of the conductors, in which case the magnetic field inside the conductors vanishes. This assumption is justified by the fact that our interest in calculating Λ is for the purpose of determining the inductance of a given structure, and inductance is of interest primarily in the ac case (i.e., time-varying currents, voltages, and fields). As we will see later in Section 7-5, the current flowing in a conductor under ac conditions is concentrated within a very thin layer on the skin of the conductor.

▶ For the parallel-wire transmission line, ac currents flow on the outer surfaces of the wires, and for the coaxial line, the current flows on the outer surface of the inner conductor and on the inner surface of the outer one (the current-carrying surfaces are those adjacent to the electric and magnetic fields present in the region between the conductors). ◀

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代入等式中的最后两个表达式后： (5.87) 并从 θ_1 积分到 θ_2，我们有

$$\mathbf{B} = \hat{\mathbf{z}}\,\frac{\mu nIa^2}{2}\int_{\theta_1}^{\theta_2}\frac{a\sec^2\theta\,d\theta}{a^3\sec^3\theta} = \hat{\mathbf{z}}\,\frac{\mu nI}{2}(\sin\theta_2 - \sin\theta_1). \tag{5.89}$$

如果螺线管长度 l 远大于其半径 a，则对于远离螺线管末端的点 P $\theta_1 \approx -90^\circ$ 和 $\theta_2 \approx 90^\circ$，在这种情况下(5.89) 减少为

$$\mathbf{B} \approx \hat{\mathbf{z}}\mu nI = \frac{\hat{\mathbf{z}}\mu NI}{l} \quad \text{(long solenoid with } l/a \gg 1\text{)}. \tag{5.90}$$

即使方程。 (5.90) 是针对螺线管中点处的 B 场导出的，它在螺线管内部的任何地方都大致有效，除了靠近端部的地方。

现在我们回到对电感的讨论，其中包括自感和互感的概念，自感代表线圈或电路与其自身的磁通链，互感涉及一个电路中由于另一个电路中的电流产生的磁场而产生的磁通链。通常，当使用术语“电感”时，意指自感。

练习 5-14：使用方程式(5.89) 以获得一个很长螺线管轴上但位于其端点的点处的 B 表达式。电磁阀端点处的 B 与中点处的 B 相比如何?

答案：端点处的 $\mathbf{B} = \hat{\mathbf{z}}(\mu NI/2l)$ 是中点处 B 的一半。（参见EM。）

5-7.2 螺线管的自感

从方程。 (5.66)，连接表面 S 的磁通量 Φ 为

$$\Phi = \int_S \mathbf{B}\cdot d\mathbf{s} \qquad \text{(Wb)}. \tag{5.91}$$

在螺线管中，其整个横截面具有近似均匀的磁场，由方程式给出。 (5.90)，连接单个环路的通量为

$$\Phi = \int_S \hat{\mathbf{z}}\left(\mu\frac{N}{l}I\right)\cdot\hat{\mathbf{z}}\,ds = \mu\frac{N}{l}IS, \tag{5.92}$$

其中S是环路的横截面积（图5-27（b））。磁通量链 Λ 定义为给定电路或导电结构的总磁通量。如果结构由具有多个环路的单个导体组成，如螺线管的情况，Λ 等于连接该结构的所有环路的通量。对于 N 匝螺线管，

$$\Lambda = N\Phi = \mu\frac{N^2}{l}IS \qquad \text{(Wb)}. \tag{5.93}$$

任何导电结构的自感定义为磁通链 Λ 与流过该结构的电流 I 之比：

$$L = \frac{\Lambda}{I} \qquad \text{(H)}. \tag{5.94}$$

电感的 SI 单位是亨利 (H)，相当于韦伯每安培 (Wb/A)。

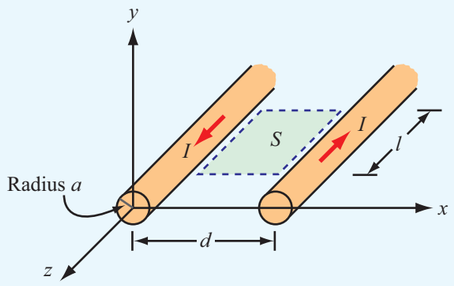
对于螺线管，使用方程式。 (5.93) 给出

$$L = \mu\frac{N^2}{l}S \qquad \textbf{(solenoid)}. \tag{5.95}$$

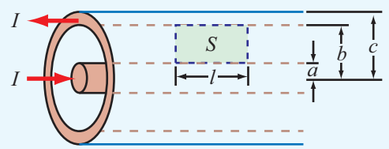
5-7.3 其他导体的自感

螺线管由横截面为 S 的单个电感器组成。另一方面，如果该结构由两个独立的导体组成，如图 5-28 中所示的平行线和同轴传输线的情况，则与任一线的长度 l 相关的磁通链 Λ 指的是通过两个导体之间的封闭表面的磁通 Φ，如图 5-28 中的阴影区域。实际上，还有一些磁通量穿过导体本身，但如果假设电流仅在导体表面流动，则可以忽略它，在这种情况下，导体内部的磁场消失。这一假设是合理的，因为我们计算 Λ 的目的是为了确定给定结构的电感，而电感主要在交流情况下（即随时间变化的电流、电压和场）而感兴趣。正如我们稍后将在第 7-5 节中看到的，交流条件下导体中流动的电流集中在导体表皮上的一个非常薄的层内。

▶ 对于平行线传输线，交流电流在电线的外表面上流动，对于同轴线，电流在内导体的外表面上流动，在外导体的内表面上流动（载流表面是与导体之间区域中存在的电场和磁场相邻的表面）。 ◀



(a) Parallel-wire transmission line



(b) Coaxial transmission line

Figure 5-28 To compute the inductance per unit length of a two-conductor transmission line, we need to determine the magnetic flux through the area S between the conductors.

For a two-conductor configuration similar to those of **Fig. 5-28**, the inductance is given by

$$L = \frac{\Lambda}{I} = \frac{\Phi}{I} = \frac{1}{I}\int_S \mathbf{B}\cdot d\mathbf{s}. \tag{5.96}$$

Example 5-8: Inductance of a Coaxial Transmission Line

Develop an expression for the inductance per unit length of a coaxial transmission line with inner and outer conductors of radii a and b (**Fig. 5-29**) and an insulating material of permeability μ.

Solution: The current I in the inner conductor generates a magnetic field $\mathbf{B}$ throughout the region between the two conductors. It is given by Eq. (5.30) as

$$\mathbf{B} = \hat{\boldsymbol{\phi}}\frac{\mu I}{2\pi r}, \tag{5.97}$$

where r is the radial distance from the axis of the coaxial line. Consider a transmission-line segment of length l as shown in **Fig. 5-29**. Because $\mathbf{B}$ is perpendicular to the planar surface S between the conductors, the flux through S is

$$\Phi = l\int_a^b B\,dr = l\int_a^b \frac{\mu I}{2\pi r}\,dr = \frac{\mu I l}{2\pi}\ln\left(\frac{b}{a}\right). \tag{5.98}$$

Using Eq. (5.96), the inductance per unit length of the coaxial transmission line is

$$L' = \frac{L}{l} = \frac{\Phi}{lI} = \frac{\mu}{2\pi}\ln\left(\frac{b}{a}\right). \tag{5.99}$$

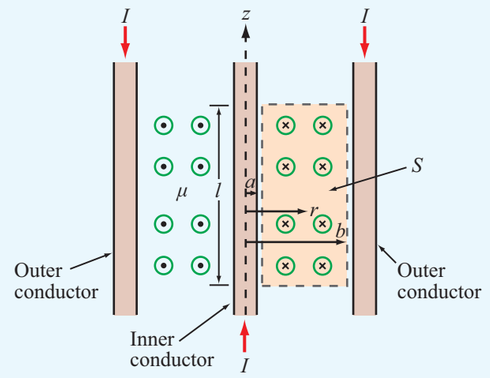
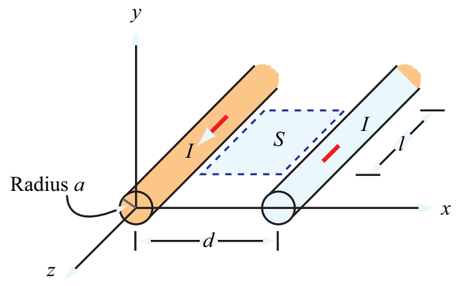


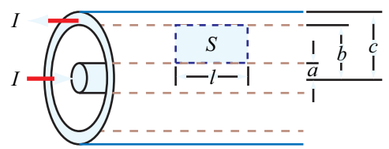
Figure 5-29 Cross-sectional view of coaxial transmission line (Example 5-8). ⊙ and ⊗ denote **H** field out of and into the page, respectively.

5-7.4 Mutual Inductance

Magnetic coupling between two different conducting structures is described in terms of the mutual inductance between them. For simplicity, consider the case of two multiturn closed loops with surfaces S_1 and S_2. Current I_1 flows through the first loop (**Fig. 5-30**), and no current flows through the second one. The magnetic field $\mathbf{B}_1$ generated by I_1 results in a flux Φ_{12}



(a) Parallel-wire transmission line



(b) Coaxial transmission line

图 5-28 为了计算双导体传输线单位长度的电感，我们需要确定通过导体之间面积 S 的磁通量。

对于类似于图 5-28 的双导体配置，电感由下式给出

$$L = \frac{\Lambda}{I} = \frac{\Phi}{I} = \frac{1}{I}\int_S \mathbf{B}\cdot d\mathbf{s}. \tag{5.96}$$

例 5-8： 同轴传输线的电感

建立同轴传输线每单位长度电感的表达式，该同轴传输线具有半径为 a 和 b 的内导体和外导体（图 5-29）以及磁导率为 μ 的绝缘材料。

解：内导体中的电流 I 在两者之间的区域产生磁场 B

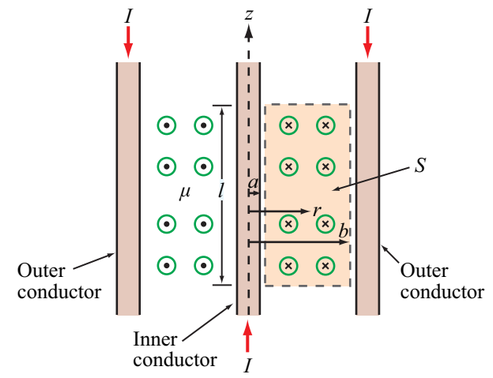


图5-29 同轴传输线的横截面图（示例5-8）。和 分别表示页面外的H 场和页面内的 H 场。

导体。它由等式给出。 (5.30) 作为

$$\mathbf{B} = \hat{\boldsymbol{\phi}}\frac{\mu I}{2\pi r}, \tag{5.97}$$

其中 r 是距同轴线的径向距离。考虑长度为 l 的传输线段，如图 5-29 所示。因为 B 垂直于导体之间的平面 S，所以通过 S 的通量为

$$\Phi = l\int_a^b B\,dr = l\int_a^b \frac{\mu I}{2\pi r}\,dr = \frac{\mu I l}{2\pi}\ln\left(\frac{b}{a}\right). \tag{5.98}$$

使用方程式(5.96)，同轴传输线单位长度的电感为

$$L' = \frac{L}{l} = \frac{\Phi}{lI} = \frac{\mu}{2\pi}\ln\left(\frac{b}{a}\right). \tag{5.99}$$

5-7.4 互感

两个不同导电结构之间的磁耦合用它们之间的互感来描述。为简单起见，考虑具有曲面 S_1 和 S_2 的两个多匝闭环的情况。电流 I_1 流过第一个环路（图 5-30），第二个环路没有电流流过。 I_1 产生的磁场 B_1 产生磁通 Φ_{12}

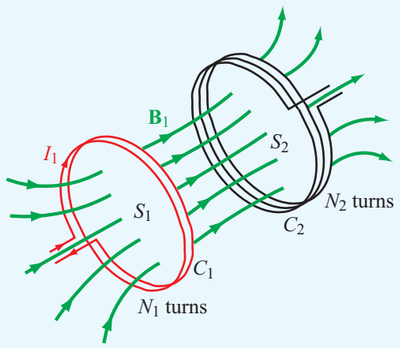


Figure 5-30 Magnetic field lines generated by current I_1 in loop 1 linking surface S_2 of loop 2.

through loop 2 given by

$$\Phi_{12} = \int_{S_2} \mathbf{B}_1 \cdot d\mathbf{s}, \qquad (5.100)$$

and if loop 2 consists of N_2 turns all coupled by $\mathbf{B}_1$ in exactly the same way, then the total magnetic flux linkage through loop 2 is

$$\Lambda_{12} = N_2\Phi_{12} = N_2 \int_{S_2} \mathbf{B}_1 \cdot d\mathbf{s}. \qquad (5.101)$$

The mutual inductance associated with this magnetic coupling is given by

$$L_{12} = \frac{\Lambda_{12}}{I_1} = \frac{N_2}{I_1} \int_{S_2} \mathbf{B}_1 \cdot d\mathbf{s} \quad \text{(H)}. \qquad (5.102)$$

Mutual inductance is important in transformers (as discussed in Chapter 6) wherein the windings of two or more circuits share a common magnetic core, as illustrated by the toroidal arrangement shown in **Fig. 5-31**.

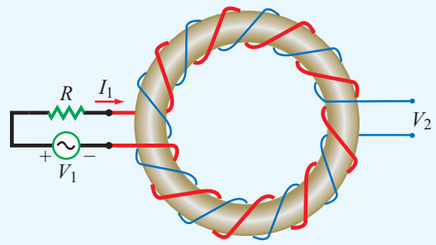


Figure 5-31 Toroidal coil with two windings used as a transformer.

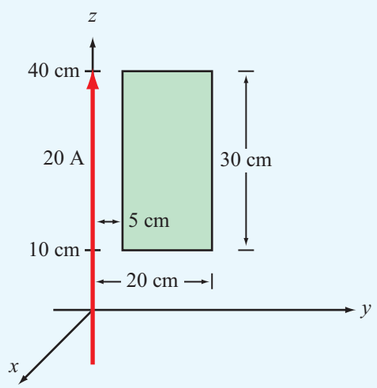


Figure 5-32 Mutual inductance between two conductors (Example 5-9).

Example 5-9: Mutual Inductance

The rectangular conducting loop shown in **Fig. 5-32** is coplanar with the long, straight wire carrying current I. Determine the mutual inductance between the wire and the loop.

Solution: According to Eq. (5.102), the mutual inductance is given by

$$L_{12} = \frac{N_2}{I_1} \int_{S_2} \mathbf{B}_1 \cdot d\mathbf{s}.$$

In the present case, $I_1 = I$ is the current carried by the first conductor (straight wire), S_2 is the cross-sectional area of the second conductor (rectangular loop), and $\mathbf{B}_1$ is the magnetic field due to the first conductor (wire). Also, $N_2 = 1$ because the loop has a single turn. In the right-hand side of the y–z plane, the magnetic field due to current I is into the page ($-\hat{\mathbf{x}}$ direction) and given by

$$\mathbf{B} = -\hat{\mathbf{x}}\,\frac{\mu_0 I}{2\pi y},$$

where y is the distance away from the wire. The expression is identical with that given by Eq. (5.97), after replacing $\hat{\boldsymbol{\phi}}$ with $-\hat{\mathbf{x}}$ and r with y. To compute the magnetic flux passing through the loop (into the page), we need to define the differential area

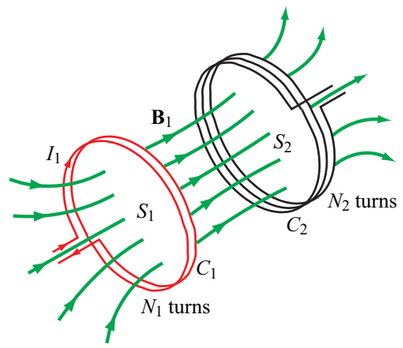


图 5-30 环路 1 中的电流 I_1 连接环路 2 的表面 S_2 时产生的磁场线。

通过由下式给出的循环 2

$$\Phi_{12} = \int_{S_2} \mathbf{B}_1 \cdot d\mathbf{s}, \qquad (5.100)$$

如果环路 2 由 N_2 匝组成，所有匝数均以完全相同的方式通过 B_1 耦合，则通过环路 2 的总磁通链为

$$\Lambda_{12} = N_2\Phi_{12} = N_2 \int_{S_2} \mathbf{B}_1 \cdot d\mathbf{s}. \qquad (5.101)$$

与该磁耦合相关的互感由下式给出

$$L_{12} = \frac{\Lambda_{12}}{I_1} = \frac{N_2}{I_1} \int_{S_2} \mathbf{B}_1 \cdot d\mathbf{s} \quad \text{(H)}. \qquad (5.102)$$

互感在变压器中很重要（如第 6 章所述），其中两个或多个电路的绕组共享一个公共磁芯，如图 5-31 所示的环形布置所示。

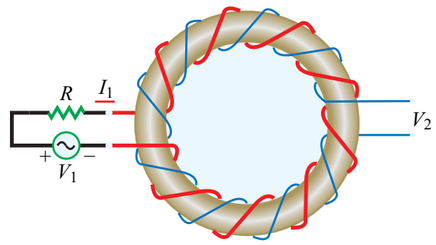


图 5-31 用作变压器的具有两个绕组的环形线圈。

例 5-9： 互感

图 5-32 所示的矩形导电环与承载电流 I 的长直导线共面。确定导线与环之间的互感。

解：根据方程(5.102)，互感由下式给出

$$L_{12} = \frac{N_2}{I_1} \int_{S_2} \mathbf{B}_1 \cdot d\mathbf{s}.$$

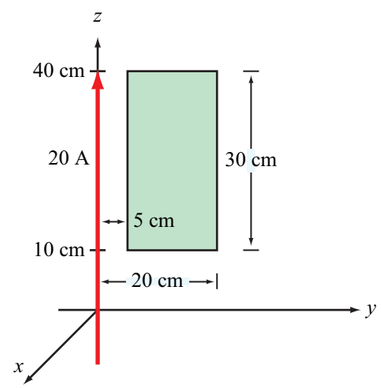


图 5-32 两个导体之间的互感（示例 5-9）。

在本例中，$I_1 = I$ 是第一导体（直线）承载的电流，S_2 是第二导体（矩形环）的横截面积，B_1 是第一导体（电线）产生的磁场。另外，$N_2 = 1$ 因为循环只有一匝。在 y-z 平面的右侧，电流 I 产生的磁场进入页面（$-\hat{\mathbf{x}}$ 方向），由下式给出

$$\mathbf{B} = -\hat{\mathbf{x}}\,\frac{\mu_0 I}{2\pi y},$$

其中 y 是距电线的距离。该表达式与式（1）给出的表达式相同。（5.97），将 $\hat{\boldsymbol{\phi}}$ 替换为 $-\hat{\mathbf{x}}$ 并将 r 替换为 y 后。为了计算穿过线圈（进入页面）的磁通量，我们需要定义微分面积